

500 Hardbacks, highest quality paper to ensure preservation for collectors, numbered 1 - 500 individually. Numbers 1 through 10 are reserved to raise money for charity or for tennis institutions, so I took [#500](#). It has nearly 200,000 words, with something for parents, coaches and tennis fans. It is the history of instruction. This was a group effort; I spoke with over 1000 coaches over the 18 years of research, and reviews are better than I dared hope for.

Parents and grassroots coaches will benefit the most. First come, first serve. If you want a specific number, ask. Free shipping if purchased in US by Jan 31st. Chuck Tomlin wrote two chapters introducing the Congruent Tennis Model after I objectively identified him as the most innovative tennis instructor of the 21st century. After discussions with Chuck, Johan Kriek, a two-time major winner, wrote the foreword. Our eminent Tennis Historian Richard Hillway wrote an article just for this book.

Lots of lost tennis history that proves today's best players follow concepts known and taught a century ago; only today, they do it with more amplitude and intensity. The buggy whip forehand used by Nadal was taught in a 1904 book better than today's "experts" IMHO. The lost black tennis history uncovered shows that it likely was a black female coach who first introduced Brandi and Richard Williams to teach Venus and Serena their open stances and powerful style. It is arguable the first modern tennis coach may have been the first woman to graduate from Yale and Harvard when she wrote a book in 1894. The photos on the cover and in the book tell their own story.

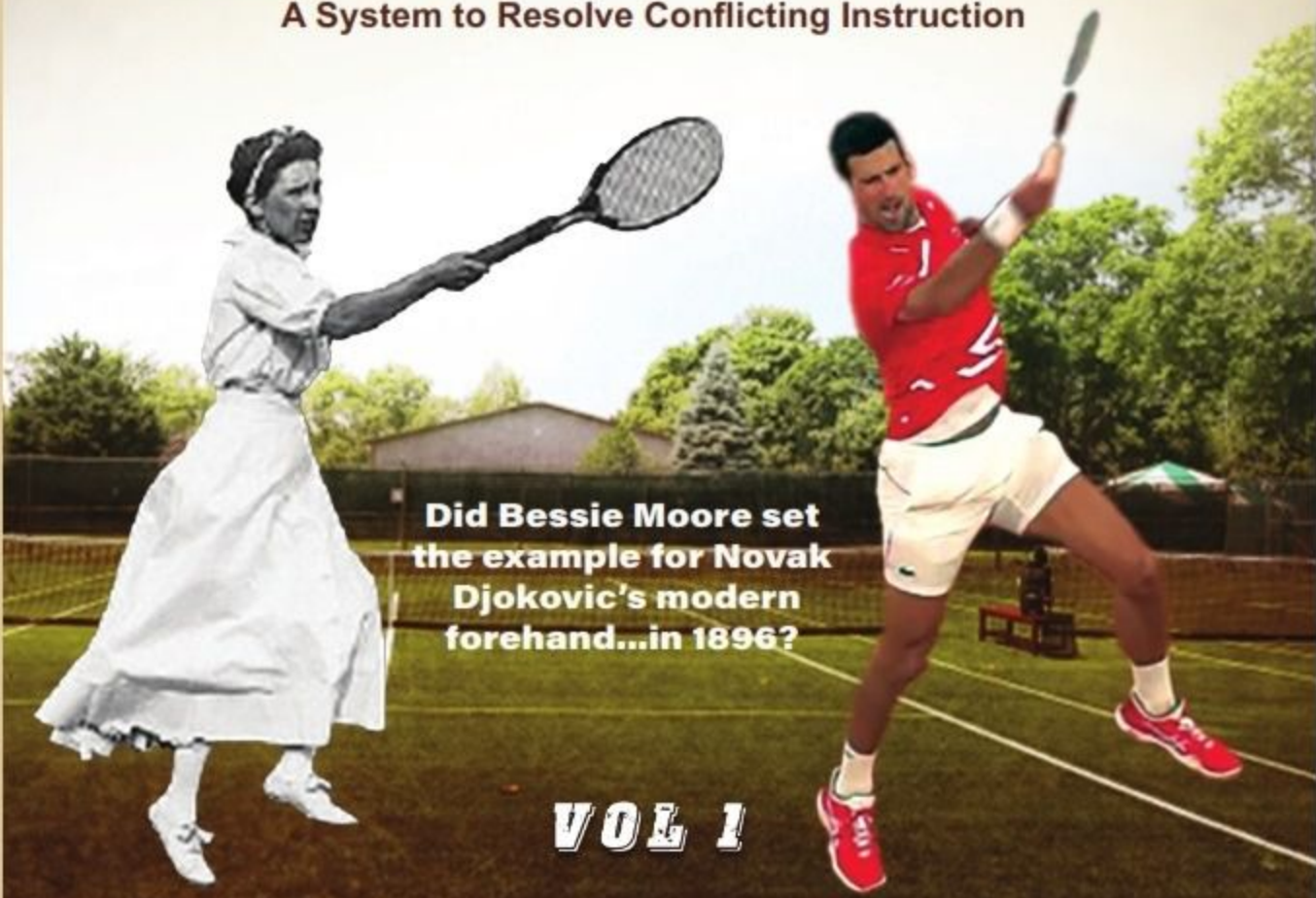
\$75 stateside shipping (free through Jan 31). It's 9" x 12" in this special edition high gloss hardback. The blank page at the front of the book is signed by myself and/or Chuck Tomlin, and we will customize the signatures per your request, or you may leave it blank. Venmo: @john-carpenter-107 or Zell: 805-200-6106 (my cell) or PayPal: [mtmcoach@hotmail.com](mailto:mtmcoach@hotmail.com) or contact me at my email [mtmcoach@hotmail.com](mailto:mtmcoach@hotmail.com) or my cell 805-200-6106 if questions. WhatsApp is 805-200-6106.

## Below Is a Preview of My Book:

# Modern Tennis Instruction

## A Historical Guide to Play Your Best Tennis

Foreword by 2-time Major Champion Johan Kriek  
Featuring the Congruent Tennis Model by Chuck Tomlin  
A System to Resolve Conflicting Instruction



Did Bessie Moore set  
the example for Novak  
Djokovic's modern  
forehand...in 1896?

**VOL 1**

**BY JOHN B. CARPENTER**

**PTR PRO & USPTA ELITE PRO**

**Limited  
Collectors' Edition**

# **MODERN TENNIS INSTRUCTION**

**A HISTORICAL GUIDE TO PLAY YOUR BEST TENNIS**  
**Volume 1: 1874-1907**

**by John B. Carpenter**

**Contributing writers: Chuck Tomlin  
& Richard Hillway**



**Elisabeth Moore, opposite Novak Djokovic on the front cover. To launch her body into the air, she coiled into her hips, hitting across and through the ball with angular momentum as top pros do today. In 1896, she won the first of four US Singles Championships.**

**Limited Edition Number**

**500/500**



## Bios



**John Carpenter** is an academic “geek” who grew up in East St. Louis, Illinois collecting baseball cards, comics, and tennis books. At age 10, a doctor told him he would likely top out at 5’ 7”, limiting his sports options. Inspired by a local, scrawny older kid, Jimmy Connors, John’s wanted to play professional tennis. At age 15, coaches told John to stop jumping in the air and to quit hitting open stance forehands. As a result, his game declined. John dreamt of being a master coach, collecting hundreds of tennis books and memorizing tenets of the acclaimed “experts” from Tilden to Braden. In 2003, he tested a contrarian method and suddenly had a waiting list of students. In 2004, he was asked to partner with a renowned international coach, Oscar Wegner, in reforming tennis instruction. In 2005, John wrote 100 pages of this book while head pro of the Dwight Davis Tennis Center in St. Louis. By 2019, he began editing over 2000 typewritten pages. In 2007, Oscar and John formed the Modern Tennis Methodology Coaching Association (MTMCA) to promote and identify what they felt was the best way to start beginners. He and Wegner challenged the USTA to a public debate on the best way to start beginners. The offer was declined. In 2016, John’s research determined that the next cutting-edge instruction frontier lay with a contrarian Atlanta coach. As a tennis instructional historian, John is currently a PTR Pro, is USPTA Elite Pro certified, and takes pride in developing juniors and adults in Atlanta.



**Chuck Tomlin**, PTR certified, contributes two instructional chapters featuring the “Fade vs. Draw” paradigm that culminated in the “Shot Matching Matrix.” A former military pilot & instructor for the US Navy Test Pilot School, and a Black Belt, he made Auburn’s varsity football team as a true freshman. After taking up coaching in 1992, he started writing the USTA with his new ideas in 1996. His tactical aviation and engineering background led him to believe tennis was missing vital fundamental concepts. While flying as an International Airline Pilot, he began to coach part-time. Looking for deeper answers, he met with and analyzed instruction with Rick Macci, Nick Bollettieri, Dennis Van Der Meer, Oscar Wegner, and Vic Braden. He developed a unique system of charting matches that focused on the missing key concepts tennis analytics previously did not measure. By 2002 he developed the “Serve Code” to explain the biggest serves. In 2007, he designed a new cognitive tactics paradigm, “Smart Targets,” a simplified visual manner to illustrate optimized hitting lanes. He shared the concept of “Dual Objectives” to deal with the dichotomy concerning details required of “coach to coach (C2C)” vs. the simplicity needed for “coach to player (C2P) speak.” He developed the Congruent Tennis Model (CTM) embracing the best aspects of widely acclaimed teaching and playing styles. In 2016, he invited John Carpenter to help finish testing the CTM with larger groups to bridge the gap between different teaching styles with a Higher Resolution approach. Coaches and pros who have reviewed the CTM attest to the benefits it could have added to their play, as well as how it resolves many conflicting issues.



**Richard Hillway** is an international tennis historian who co-authored **The Birth of Lawn Tennis** in 2018. He published over 125 tennis history articles, many in *The Tennis Collector* and *The Tennis Historian* in England, and in the *Journal of the Tennis Collectors of America*. Owner of over 2,500 tennis books and periodicals, he is considered an expert on the origins of lawn tennis, its inventor Major Walter Wingfield, and Bill Tilden. A teaching pro for 55 years, he won 46 high school state team titles as head or assistant coach in Denver and Colorado Springs. He also served as Tennis Director for 28 years at the Country Club of Colorado. Self-taught, he hit “reverse” (bolo) and open stance forehands in the 1950s. He played number 1 singles for the University of Colorado from 1963 to 1965 and was chosen Outstanding Senior Athlete Scholar. He attended the University of South Wales in Australia on a Rotary International Fellowship in 1967, winning the All Australian University tournament. He ranked number 1 in 1966 and 1970 in men’s open singles in the six-state Intermountain section. The most exciting match of his career was losing in the 1966 Colorado State Open to 20-year-old Jaime Fillol, (later world No. 14) 19-17, in the third set. He founded the Colorado Tennis Hall of Fame in 2000. **American Colossus**, the 2018 biography of Bill Tilden, is dedicated to Hillway.



# A Tribute to the Tennis Historians on Whose Shoulders I Stand

By John Carpenter

One of the challenges of documenting an accurate history of instruction is that so much of tennis history has somehow been distorted by misremembering, lack of proper record-keeping, the invention of false stories, or simply becoming lost. My historical editor, Richard Hillway, was fortunate to have the world's leading historians as private mentors. Frank Phelps (d.2010) was America's leading tennis historian for decades, the one Bud Collins totally trusted. George Alexander (d.2008) was the most significant historian of the life of tennis inventor, Major Walter Wingfield. Alan Little (d.2017) from whom Richard learned about everything Wimbledon, was the Honorary Wimbledon Librarian for forty years from 1977 to 2017. Two more mentors were Mrs. Patricia Barry (d.2015), who spent decades examining the family of Mary Outerbridge, and Attila Szabo, the world's foremost expert on tennis books. New York's Geoff Felder spent three decades doing nineteenth-century tennis research and helped Frank Phelps and Richard Hillway research their various projects. Also involved with correcting tennis history are England's Bob Everitt, who co-authored **The Birth Of Lawn Tennis** (2018) with Hillway, and Richard Jones, who owns The Tennis Gallery bookstore in Wimbledon. Jones works with Bob Everitt on *The Tennis Historian* magazine.

The pictures on this page are three images as they appeared in early 2023 about the first female, May Sutton Bundy, elected to the International Tennis Hall of Fame in 1956. All three contained errors. Notice this Wikipedia entry about May Godfray Sutton states she won her first US Championship at the age of 16.

From Wikipedia, the free encyclopedia

**May Godfrey Sutton** (September 25, 1887 – October 4, 1975) was an American tennis player who was active during the first decades of the 20th century. At age 16 she won the singles title at the U.S. National Championships and in 1905 she became the first American player to win the singles title at Wimbledon.

Bundy, elected to the International Tennis Hall of Fame in 1956. All three contained errors. Notice this Wikipedia entry about May Godfray Sutton states she won her first US Championship at the age of 16. She won it in June 1904 at the age of 17.

It takes a lot of time to fact-check these various claims. This 2023 Tennis Hall of Fame bio states May was 40 years old when she played the 1925 Wightman Cup. To fact-check this bio, I looked up records of the Wightman Cup. She was only 38 years old at that time. When I brought it to Hillway's attention, he noticed something else. It was not true that her grandson, Brian Cheney, was highly ranked in the 1940s because in the 1990s, he was ranked number one in the over-40s singles and doubles divisions. The time frame is off by half a century.

Unfortunately, such mistakes can be found everywhere in books, articles, and especially on the internet, Wikipedia, and even at the ITHOF. Tennis historians expend a lot of effort to find evidence to correct so many errors, often simple ones. Frank Phelps corrected many errors found in the Tennis Hall of Fame bios, but Hillway has carried on the task since then. Notice May's tombstone. It has the wrong date of birth, making it appear that she died at 88.



Years later, Frank Phelps tried to find a record of May Sutton's birth in America, but was unsuccessful. He wrote to Alan Little in England, who discovered May's birth certificate. It showed she was born in Plymouth, England, before her parents moved the family to America when May was a young child. May Sutton Bundy was born in 1886, not in 1887, and died at age 89. By then, the error had quickly spread to printed media and internet media. Also, her middle name is Godfray, not Godfrey.



# Table of Contents

**Foreword** - by Johan Kriek: Bringing Artistry Back to Tennis Begins with the Right Foundation

**Preface** - by John Carpenter: How I Became a Tennis Instruction Historian

**Intro** by Steve Kaplan: Why Coaches, Parents, and Students of Tennis Might Want to Read This Book

## **CHAPTER 1: How the Eyes Can't See what the Mind Doesn't Know. . . . . 1**

|                                                                                          |           |
|------------------------------------------------------------------------------------------|-----------|
| <b>The History of Tennis Instruction- An Evolutionary Lens to Guide Us. . . . .</b>      | <b>1</b>  |
| <b>The Seven Chapters of Volume I. . . . .</b>                                           | <b>2</b>  |
| <b>The First Book that Claimed to Teach Modern Tennis. . . . .</b>                       | <b>3</b>  |
| <b>Buggy Whip, Straight Arm, &amp; Inside-Out FHs Were Used before 1900. . . . .</b>     | <b>4</b>  |
| <b>The Origins of the Debate Over Open Versus Closed Stance Forehands . . . . .</b>      | <b>7</b>  |
| <b>Use of the Hips was Taught From Tennis' Earliest Days . . . . .</b>                   | <b>9</b>  |
| <b>How My Eyes Could Not See What My Mind Did Not Know. . . . .</b>                      | <b>11</b> |
| <b>What does "Stances are Situational" Mean? . . . . .</b>                               | <b>12</b> |
| <b>Who exposed Richard Williams to Open Stance Forehands? . . . . .</b>                  | <b>13</b> |
| <b>Do we Know the First Black Coaches to Write Instructional Books? . . . . .</b>        | <b>15</b> |
| <b>An Overview of Our Instructional History . . . . .</b>                                | <b>16</b> |
| <b>Breakthroughs in Improving Instruction: Congruency &amp; Dual Objectives. . . . .</b> | <b>17</b> |
| <b>What Makes the Tennis Establishment Resistant to Change? . . . . .</b>                | <b>19</b> |
| <b>Did we Ask the Wrong Questions or Not Ask the Right Ones? . . . . .</b>               | <b>20</b> |
| <b>How Do You Cognitively Teach Students to Hit the Rising Ball? . . . . .</b>           | <b>21</b> |
| <b>Abolish the USTA? Surely You Can't Be Serious! . . . . .</b>                          | <b>21</b> |
| <b>Importance of Paradigm Shifts in Tennis History. . . . .</b>                          | <b>22</b> |
| <b>The Ingenuity of Grassroots and Developmental Coaches. . . . .</b>                    | <b>23</b> |
| <b>The Tennis Manufactory . . . . .</b>                                                  | <b>25</b> |
| <b>The Greatest Instructional Book Never Published. . . . .</b>                          | <b>25</b> |
| <b>The Farther Backward You Look, The Farther Forward You Can See. . . . .</b>           | <b>26</b> |

## **CHAPTER 2: Did Ski Instructors Teach Tennis Better Than Tennis Coaches? . . . . . 27**

|                                                                                           |           |
|-------------------------------------------------------------------------------------------|-----------|
| <b>An "Inner Game of Tennis" Clinic Yields Coaching Insights. . . . .</b>                 | <b>27</b> |
| <b>Does Certification by Tennis Institutions Guarantee Quality Instruction? . . . . .</b> | <b>28</b> |
| <b>Why Tennis Builds Strong Bones and Strong Minds. . . . .</b>                           | <b>30</b> |
| <b>A Child Found On A Tennis Court will not be Found in Juvenile Court. . . . .</b>       | <b>31</b> |

## **CHAPTER 3: To What extent Can Science Help Us Understand Good Form and How to Improve Our Instincts? . . . . . 36**

|                                                                                          |           |
|------------------------------------------------------------------------------------------|-----------|
| <b>Scientific Claims versus Outcomes On-Court: Points of Intersection. . . . .</b>       | <b>36</b> |
| <b>The Role of the Wrist in Tennis Strokes-Do We Understand It . . . . .</b>             | <b>37</b> |
| <b>The Coach Who Applied Time and Motion Principles to Tennis Biomechanics . . . . .</b> | <b>38</b> |



|                                                                                                                                                          |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| The Hips-Shoulders-Hands Sequence. . . . .                                                                                                               | 39     |
| The Coach who Brought Computerized Biomechanics Comes to Tennis. . . . .                                                                                 | 40     |
| What is Good Form?.. . . .                                                                                                                               | 42     |
| What Can We Learn From the History of Biomechanics? . . . . .                                                                                            | 44     |
| To What Degree can Video Analysis Improve Tennis Instruction?.. . . .                                                                                    | 45     |
| How Much is Racquet Head Acceleration Needed for Power and Spin?.. . . .                                                                                 | 47     |
| Intro to Chapter 4: The Chronology & History of Tennis Instruction. . . . .                                                                              | 48     |
| <br>CHAPTER 4: The History of Instruction Timeline Begins- Did Players More Than a<br>Century Ago Use the Grips, Stances, and Styles of Today? . . . . . | <br>50 |
| Two Important Books Capture the Significance of Tennis History . . . . .                                                                                 | 50     |
| The Origins of the First Tennis Vocabulary . . . . .                                                                                                     | 53     |
| How Modern was the First Instructional Article in 1876? . . . . .                                                                                        | 53     |
| The First Tennis Instructional Book - Hit Through or Across the Ball? . . . . .                                                                          | 55     |
| How Tennis Grips Got Their Names... . . . .                                                                                                              | 56     |
| The USTA's Predecessors. . . . .                                                                                                                         | 59     |
| Were Straight arm Forehands used in the Nineteenth Century? . . . . .                                                                                    | 61     |
| The Original Two Forehand Concept. . . . .                                                                                                               | 62     |
| The Debate Begins Between Neutral and Open Stance Forehands. . . . .                                                                                     | 63     |
| The Case is Made in Favor of the Neutral Stance Forehand.. . . .                                                                                         | 65     |
| The Plain Face Racquet Forehand Drive – Fade Topspin? .. . . .                                                                                           | 66     |
| The Fabulous Lottie Dodd.. . . .                                                                                                                         | 67     |
| Slocum's View on Good Form . . . . .                                                                                                                     | 69     |
| American Women Began Best of Five Set Matches in 1891 . . . . .                                                                                          | 70     |
| The First Identified Case of Coaching Plagiarism. . . . .                                                                                                | 70     |
| Was This Female Instructor a Hundred Years Ahead of Her Time? . . . . .                                                                                  | 72     |
| An 1894 Drill to Increase Racquet Head Acceleration. . . . .                                                                                             | 73     |
| It is Easy to Learn Well as Badly. . . . .                                                                                                               | 76     |
| Origin of the “Forehand Power Fade: aka “The Smith.” . . . .                                                                                             | 76     |
| British Men and Women Teach the “New” Neutral Stance Forehand. . . . .                                                                                   | 79     |
| Origins of the Davis Cup. . . . .                                                                                                                        | 81     |
| Does Learning to Play on Clay Produce Better Players? . . . . .                                                                                          | 81     |
| The “Forcing Play” and 19 <sup>th</sup> Century Tennis Analytics. . . . .                                                                                | 82     |
| All the Best Women by 1900 Got Better by Playing Men. . . . .                                                                                            | 84     |
| Ralph Waldo Emerson's Wisdom Applied to Tennis Instruction. . . . .                                                                                      | 86     |
| Golden Gate Park's Eight Red Clay Courts. . . . .                                                                                                        | 87     |
| <i>The Ghosts of Lawn Tennis Past</i> by Eaves, Lake and Cowdrey. . . . .                                                                                | 88     |
| The First Debate on How to Apply Spin to the Ball. . . . .                                                                                               | 89     |
| The Americans Are Off to A Running Start! Somebody STOP Them! . . . . .                                                                                  | 95     |
| Was P.A. Vaile the Godfather of Modern Tennis Instruction? . . . . .                                                                                     | 97     |
| The Importance of Rotation: Twist, Lift, Cut Top, Duck, Side, & Chop. . . . .                                                                            | 98     |
| Keep Your Eyes on the Part of the Ball You Want to Hit. . . . .                                                                                          | 101    |
| Vaile Warned About the Many Fads of Tennis . . . . .                                                                                                     | 103    |



|                                                                                         |            |
|-----------------------------------------------------------------------------------------|------------|
| <b>The Two Forehand Theory in 1904: Flatter Balls vs. Spinning Balls. . . . .</b>       | <b>104</b> |
| <b>Did Vaile Forecast the Most Effective Shot in Modern Tennis? . . . . .</b>           | <b>105</b> |
| <b>The American Coach Who Wrote on Instruction for 50 Years. . . . .</b>                | <b>109</b> |
| <b>Do Deep Shots Limit the Effectiveness of the Attack? . . . . .</b>                   | <b>110</b> |
| <b>Paret Taught the Buggy Whip Forehand Used by All Players Today. . . . .</b>          | <b>113</b> |
| <b>The Tennis Manufactory – The Golden Gate Junior Tennis Club.. . . .</b>              | <b>116</b> |
| <b>Who was the Master Developmental Coach behind GGJTC? . . . . .</b>                   | <b>118</b> |
| <b>It Takes a Sutton to Beat a Sutton. . . . .</b>                                      | <b>119</b> |
| <b>What Isaac Newton Learned from Watching Tennis . . . . .</b>                         | <b>124</b> |
| <b>Did P.A. Vaile Revolutionize Golf and Tennis Both? . . . . .</b>                     | <b>126</b> |
| <b>Case Study: When Does Criticizing Other Coaches Cross the Line? . . . . .</b>        | <b>130</b> |
| <b>Payn’s Attack on Vaile’s First Instructional Book. . . . .</b>                       | <b>132</b> |
| <b>Payn’s Attack on Vaile’s Second Instructional Book. . . . .</b>                      | <b>132</b> |
| <b>Payn’s Attack on Vaile’s Third Instructional Book . . . . .</b>                      | <b>134</b> |
| <b>S. Wallis Merrihew Published the Best Tennis Magazine for Four Decades . . . . .</b> | <b>135</b> |
| <b>Revelations from Volume II that Give Context to Volume . . . . .</b>                 | <b>137</b> |
| <b>We Have a Lot of Tennis History Left to Correct. . . . .</b>                         | <b>138</b> |
| <b>Did Tilden Study Other Books and Tennis Theories . . . . .</b>                       | <b>139</b> |
| <b>The Most Unique Instructional Book of the 20<sup>th</sup> Century . . . . .</b>      | <b>140</b> |
| <b>Vaile’s Last Tennis Book . . . . .</b>                                               | <b>144</b> |
| <b>How Bill Tilden’s First ALT Article Exposed His Weakness . . . . .</b>               | <b>146</b> |
| <b>Tilden on What is a Great Tennis Player . . . . .</b>                                | <b>148</b> |
| <b>The First Film Analysis of Top Players . . . . .</b>                                 | <b>149</b> |
| <b>Paret’s Introduction of Biomechanics to Tennis . . . . .</b>                         | <b>149</b> |
| <b><i>Revolutionarytennis.com</i> – In Memory of Mark Papas . . . . .</b>               | <b>150</b> |
| <b>Who was the First Tennis Scientist?. . . . .</b>                                     | <b>153</b> |

## **Chapter 5: Improving Vocabulary with the Congruent Tennis Model (CTM)**

by *Chuck Tomlin*

|                                                                     |            |
|---------------------------------------------------------------------|------------|
| <b>Keeping Instruction Simple with a Higher Resolution. . . . .</b> | <b>154</b> |
| <b>Technique - the Key to Consistent Execution. . . . .</b>         | <b>155</b> |
| <b>Key Concept Number 1 – Stable Platform. . . . .</b>              | <b>156</b> |
| <b>Key Concept Number 2 – Position to Win. . . . .</b>              | <b>156</b> |
| <b>Key Concept Number 3 – Power Isn’t Everything. . . . .</b>       | <b>157</b> |
| <b>3 Instructional Processes to Improve Results. . . . .</b>        | <b>157</b> |
| <b>Bringing Synthesis to Techniques. . . . .</b>                    | <b>158</b> |
| <b>Technical Flaws. . . . .</b>                                     | <b>161</b> |
| <b>3 Perspectives on Tactics. . . . .</b>                           | <b>163</b> |
| <b>Three Main Phases of Play. . . . .</b>                           | <b>165</b> |
| <b>Attacking Strategies. . . . .</b>                                | <b>165</b> |
| <b>Rally Shots versus Forcing Attacks. . . . .</b>                  | <b>166</b> |
| <b>Serves and Returns. . . . .</b>                                  | <b>167</b> |
| <b>The Four Steps of the Serve Code. . . . .</b>                    | <b>167</b> |
| <b>Returning Serves or Any Powerful Shot. . . . .</b>               | <b>170</b> |



|                              |     |
|------------------------------|-----|
| First Strike Tennis. . . . . | 170 |
|------------------------------|-----|

## Chapter 6: Teaching Tennis Through a Higher Resolution by *Chuck Tomlin*

|                                                                  |     |
|------------------------------------------------------------------|-----|
| Using an Advanced Tennis Foundation to Build Congruence. . . . . | 172 |
| Smart Target Hitting Lanes. . . . .                              | 173 |
| What is a Quality Ball? . . . . .                                | 176 |
| Bringing the Ball Down. . . . .                                  | 176 |
| Working the Ball at Contact. . . . .                             | 177 |
| Dual Objectives. . . . .                                         | 177 |
| A New Dual Objective for Shot Creation. . . . .                  | 178 |
| The Impact of Rising or Dropping. . . . .                        | 179 |
| 3 Aspects of the Contact Point. . . . .                          | 179 |
| Are Spin and Power Inversely Related? . . . . .                  | 180 |
| ATP Forehand or “Timing Enhanced Forehand” . . . . .             | 180 |
| The Slot Entry and the Slot Path. . . . .                        | 181 |
| Lag to Drag. . . . .                                             | 182 |
| The 2 Topspins Concept: To Unleash Your Natural Power . . . . .  | 182 |
| Rock, Paper, Scissors...A Simple Way to Shot Match. . . . .      | 184 |
| The Shot Matching Cycle. . . . .                                 | 184 |
| Importance of Proper Matching Feeds. . . . .                     | 186 |
| Charting to Improve Performance. . . . .                         | 186 |

## Chapter 7: How Parents, Coaches, and Players of all Levels can Navigate the Challenge of Contradictory Instruction

|                                                                                     |     |
|-------------------------------------------------------------------------------------|-----|
| Lessons Learned from the 1874 to 2023 Timeline. . . . .                             | 188 |
| Is Originality Making Something New out of Something that Already Exists? . . . . . | 189 |
| We Praise the Coach That Got Us There. . . . .                                      | 190 |
| Online Instructors and Writers. . . . .                                             | 192 |
| The Danger of Fads Are Still Present. . . . .                                       | 194 |
| In Memory of England's John Oliff. . . . .                                          | 195 |
| Does Tennis have a Pickleball Problem? . . . . .                                    | 195 |
| What does it Mean to Play Naturally? . . . . .                                      | 196 |
| Practice Off the Court – Tennis Wisdom from History . . . . .                       | 196 |
| Why Defining Beginner Fundamentals Should Be Priority Number One. . . . .           | 197 |
| The Most Innovative Coach in the Twenty First Century. . . . .                      | 198 |
| Where Are We Now and Where do we go from here? . . . . .                            | 200 |
| More Instructional History Needs to be Uncovered. . . . .                           | 201 |



## Appendixes

|                                                                                                            |     |
|------------------------------------------------------------------------------------------------------------|-----|
| A: Coach Wayne Bryan's 2012 Open Letter to the USTA.....                                                   | 204 |
| B: Coach Sidney Llewellyn's 1984 Interview from "Black Champions".....                                     | 212 |
| C: 1884 Article: "A Scientific Study of Lawn Tennis" by C. L. Dana, M.D. ....                              | 220 |
| D: Todd Martin's Keynote Speech at the 2005 US Open Tennis Teacher's Conference.....                       | 223 |
| E: Hit Through or Across the Ball? A New Direction for Racket Head .....                                   | 229 |
| F: Congruenttennis.com Website: An Open-Source Instruction Platform.....                                   | 232 |
| G: Glossary of Tennis Definitions to Resolve Contradictions.....                                           | 233 |
| H: "Comparison: The Greatest Tennis Player Ever and the Grand Slam Tournaments" by<br>Richard Hillway..... | 237 |
| I: "The Best Tennis Instruction Book Never Published by a Parent/Coach".....                               | 284 |
| J: History of Tennis Instruction Quiz #1.....                                                              | 302 |

**All footnotes annotated in this book are listed at  
[www.congruenttennis.com/Vol1Footnotes](http://www.congruenttennis.com/Vol1Footnotes)**



## Foreword by Johan Kriek

### Bringing Artistry Back to Tennis Begins with the Right Foundation

During my 18-year professional tennis career, I enjoyed the rare privilege of living in a significant time period of tennis history. I had the opportunity to play on center court in many majors during an era of rapid changes in equipment technology and playing styles. Tennis is a sport of tradition, and I have great regard for its history due to the unique perspective of an ATP Tour player. I also have a fascination with tennis technique born out of my passion for coaching others to enjoy this fine sport that brought me both joy and heartbreak. It was never boring and reflecting back, it was always a thrill.

I am one of the pros who played with all four major changes in racket technology. I started my pro career in 1976 with wood rackets (Dunlop Maxply), then switched to the metal frame of the "Red Head" racquet. Next, I used a metal/carbon/plastic/foam hybrid from Rossignol. I rounded out my career with all graphite models from Donnay, Yonex and Wilson.

I have read many books on tennis history, as well as just about every popular "how to" book on its instruction. With no hesitation, I can say that this book is a masterpiece and one of the greatest ever written on "tennis history and instruction" for this amazing sport. The research that has gone into this work, along with its attentiveness to detail and a vividness of description, is unparalleled. It answers questions that before, I never thought to ask, with its keen perspective that is refreshing and eye-opening.

In the late 1970s, I played against Jimmy Connors. He had this flat forehand drive with an incredibly low, long bounce. He could "work" the shot so that when he hit it down the line as a left-hander, the ball would slide further away from his opponent. This made it more difficult to pass him cross-court on the full run. Jimmy hit what Chuck Tomlin accurately calls a "fade drive", brilliantly drawing on golf terms to perfectly describe the shots that I faced. It was uncanny how low over the net Connors hit his shots with his two-handed backhand. What once had been baffling, now makes perfect sense after reading in this book how precise a fade drive can be hit for net clearance.

Andre Agassi hit the high-contact forehand which faded away from my backhand. I also faced Dick Stockton hitting his great dying fade slice that the Congruent Tennis Model shares as one of the three daggers used for mid-court attack shots. I can picture Boris Becker hitting his signature "inside-out fade" that was so effective on the return of serve from the deuce court. John McEnroe had this tough inside-out volley that "faded." My own personal favorite shot was a running cross-court, dipping "draw" I often hit with a buggy whip finish to pass players coming to the net, the CTM 4th dagger. Pete Sampras later made this shot famous.

The artistry of yesteryear's players is fast becoming a distant memory. The powers that be have mastered the way to make grass play like French clay and hard courts to be as slow as clay. This court surface homogenization has changed tennis forever... maybe. But as this book points out, by reevaluating tennis at a higher resolution, coaches can better teach players how to shape shots, despite the increased amplitude of today's swings. Despite new technology and bigger players, many aspects of power can be overcome by a return to fundamentals to support the game's artistry.

My hope is that the CTM's emphasis on shaping shots and its four-dagger system can offset the overemphasis on power that makes so many players look like they came out of the same factory. Beginners and intermediates should learn how to shape balls to build feel and instinct, with training that can unleash their skill and athleticism.

It is a different "animal" that plays tennis today, but come to think about it, it has always been so. The stars of one time period were always a bit different from one era to the next. The sport has consistently evolved in many ways, and as this book reveals, much of what is new is also old. That is what makes the history revealed here so fascinating to me. If tennis is to regain its artistic individualism, it must begin with the true fundamentals. Using the principle of congruency as a guide to teaching tennis while keeping the feel and

instinct, also makes total sense. The more the sport changes, the more it has stayed the same in some very key ways.

This book claims the Congruent Tennis Model offers a higher resolution to teach tennis with, and I'm here to confirm that it does! This is coaching at a whole different level. I love this game and if you love it as I do, you will truly love this book.



**Johan Kriek hitting a running FH "Buggy Whip Lift and Spin Draw." Besides beating clay-court legend Guillermo Vilas in the French Open quarters, he defeated greats: Agassi, Connors (2 times), Gerulaitis (4 times), and John McEnroe (5 times).**

Kriek won two majors, was 5'7" (1.75 meters) and was known for running down any ball. Born in South Africa, he became a US citizen in 1982. He reached the quarterfinals or better in all four Grand Slam majors, winning the Australian Open titles in 1981 and 1982. He was twice quarterfinalist and once a semi-finalist at the US Open, reached the semi-finals at the French Open, the quarterfinals of Wimbledon, and the semis of the Australian after winning it twice. He gives back by running a charity that delivers clean water to disadvantaged communities around the world.



# INTRODUCTION

**By Steve Kaplan, Owner/Director Bethpage Park Tennis Center, Long Island, NY**

## **Why Coaches, Parents, and Students of Tennis Might Want to Read This Book**

John Carpenter called me from Atlanta around the time of the US Open back in 2019 BC (Before Covid), seemingly out of the blue.

He explained that he read many of the articles I had written for a local, well-regarded publication, *LI/NY Tennis Magazine* (Long Island and New York Tennis Magazine.) He had written a book on the history of tennis instruction and thought I might be open to peer-reviewing some new teaching concepts. I didn't know him, so I was skeptical, but John was so persistent and enthusiastic, I couldn't say "NO!" We arranged to meet at my club, Bethpage Park Tennis Center on Long Island.

He brought Chuck Tomlin with him and introduced him as "my equal in tennis knowledge." They talked for an hour about a new paradigm for tennis instruction. Now I was even more skeptical and wondered what these guys were trying to sell me? I listened politely to what they had to say and then was passively dismissive.

And for several years, I gave this meeting little thought.

When I reflect back on this encounter, I wonder to myself, why was I so resistant to their novel ideas? After all, I had always thought of myself as the ultimate anti-tennis establishment coach. Why didn't I even want to give them a chance? This brings me to the present, in what pretentious intellectuals call, "a moment of ego clarity."

Like many tennis coaches, I might have as my epitaph, "Often Wrong but Never in Doubt." I preach humility to my students every day, but coaching is empowering, and such power corrupts at times. My epiphany, however, gives me a greater understanding of why John and Chuck have found the need to write this book and why the tennis industry really needs this book.

For many years, the foundation of tennis information that most fans read and hear has been built and credited to the work of former top players. We read their work in tennis publications and watch them on tennis broadcasts. The industry gives these former stars credibility in a kind of, "If you can dodge a wrench, you can dodge a ball," inference logic which is a funny line in a movie, but less so when it is the foundation for choosing the voices of an industry. Therein lies the problem. Great players know what they know, and they know how to play great, but not always why they played great. Great players see the big picture of how to win, and I believe that they see tennis as a grand scheme process.

These great players are the classic examples of what the Greek poet Archilochus talked about in his fable about the Fox and the Hedgehog. The Fox is clever, quick, and always on the move, gathering information and slickly using it to extricate itself from crisis. The Hedgehog, on the other hand, has a coat of spikes, so he simply hunkers down and moves ahead. While the Fox knows many subtle things, the Hedgehog only needs to see one big thing.

We learn like a Fox, but at the highest level, we play like a Hedgehog because we have already learned. Armed with this knowledge, we just put our heads down and bully ahead, focused only on

the prize. The tennis teaching industry has been built on the broad shoulders of Hedgehogs, but it desperately needs the wisdom of The Fox.

Enter John Carpenter and Chuck Tomlin.

John seeks answers in history, which is a nearly forgotten ideal, but as Robert Heinlein said, "*A generation which ignores history has no past-and no future.*" Tennis instruction is steeped in history, but why should I, as a coach, care about what someone with tweed knickers and a wood racket, strung with someone's house pet named "Whiskers" said about how to hit a running forehand 100 years ago? (Sorry, John, I know it's cow intestine, not "catgut.")

As John knows so well, the answer is that "*Past is prologue*," as Antonio in Shakespeare's, *The Tempest* first noticed. Everything that has taken place in the history of tennis instruction is a preparation for the opportunities to come, and don't we as coaches want to be cutting edge in our understanding of our craft? This is where Chuck's model for tennis congruence comes in because it is the future of tennis instruction and will stand the test of time because it is based on immutable laws of the physical world. Many coaches already know what Chuck will present, but knowing is not the same as understanding and is a far cry from mastery.

Coach Tomlin understands that tennis is a 3-D sport that has been previously explained in 2-D words using one-dimensional techniques. His gift is in his ability to organize a system and a vocabulary that removes barriers and take tennis instruction from the darkness of analog explanations into the light of higher-resolution understanding.

John and Chuck were born to undertake this challenge and have dedicated their professional lives to satisfying their curiosity which is the defining feature of leadership.

They have no ego issues because they not only mentor as coaches, but they reverse mentor and seek out the mentorship of others wherever they can find it, as you will read in this book.

John and Chuck answer more questions in this book than most tennis "experts" ever thought to ask and raise more questions than they answer in the process. And that is the point of this work because if you are a coach, you are a student first, always asking why and always searching for answers and truth.

I do not doubt that the curiosity of these professionals will inspire them to continue to seek more truth and will inspire the curiosity and interest of all coaches, as they have inspired me.

*Steve Kaplan is the owner and managing director of Bethpage Park Tennis Center, as well as director emeritus of Lacoste Academy for New York City Parks Foundation. He is also executive director and founder of Serve & Return Inc., with the motto, "All it takes is a Racquet and a Dream." Steve has coached more than 1,100 nationally ranked junior players, 16 New York State high school champions, two NCAA Division 1 Singles Champions, and numerous highly ranked touring professionals. Many of the students Steve has closely mentored have gone on to achieve great success as prominent members of the New York financial community and in other prestigious professions. In 2017, the USTA awarded Steve the Hy Zausner Lifetime Achievement Award. He may be reached by e-mail at StevenKaplan2@gmail.com.*



# Chapter 1

## How the Eyes Can't See What the Mind Doesn't Know

*"Science is belief in the ignorance of experts."*

—Richard Feynman, Nobel Prize winner in Physics

### The History of Tennis Instruction- An Evolutionary Lens to Guide Us

What does the Richard Feynman quote above have to do with tennis? A great deal indeed, if one examines the evidence presented in the 150-year historical timeline regarding tennis instruction, particularly if you consider how it is best played and taught by those acknowledged as experts. When I typed an initial hundred pages of this book by the end of 2005, I did not realize it would take the next fifteen years of researching, documenting, and testing every method I could find in tennis literature before it was ready to edit.

How does one determine what good tennis instruction is? How can we identify the best developmental coaches to start our children off correctly? What are the key fundamentals that set an advanced tennis foundation for each beginner? How do parents, players and coaches avoid the plague of tennis: contradictory instruction. This ambitious book asks tough questions and identifies potential solutions.

Chapter 4 begins the first year-by-year timeline of tennis instruction since 1874. Though certified as a PTR Pro and USPTA Elite Pro, I had no idea how many valuable insights our instructional history had to offer. Once I finished documenting who was teaching what and when from 1874 to the current year, I cut the timeline into three volumes due to length. This first edition of Volume I covers the years 1874 to 1907.

It is essential to have a written heritage to learn from our coaching ancestors. Studying how pioneers debated the best teaching methods allows us to see which patterns have emerged. It points to new directions to resolve today's instructional conflicts, particularly between the grassroots coaches and the three major US tennis institutions, the USTA, USPTA, and PTR. We now have context for the origin of specific tips. Which insights became dogma, why were specific tips later discarded, and which tenets stood the test of time? Many coaches appeared surprised to discover coaches taught the "drag the racquet" concept in tennis' earliest days. Was it possible that the techniques of today's "modern tennis" were introduced by instructors over a century ago, differing only in degree of intensity? Who were the seminal pioneers? New facts and evidence in the timeline will likely surprise those "in the know" of tennis' past.

In 1979, Allen Fox inspired me to collect books and fill notebooks with different tips when he identified the main problem with teaching tennis, the curse of contradictory instruction. Coach Fox's words ring true today:

*Unfortunately, tennis is still something of a black art. The game abounds with charlatans and people peddling their own pet theories. In many cases, advice from two tennis pros will be directly contradictory. It is a nasty position to find yourself in after approaching both in a state of ignorance. If you knew enough to decide which was right, you probably wouldn't be taking lessons in the first place. Although it might not be much consolation, it should be known that the problem is not restricted to beginners. Many of the top players in the world are equally confused by a myriad of conflicting advice when they suddenly develop trouble with their serve swing or backhand. I wish there was an easy answer, but there isn't.<sup>1</sup>*





**Want to watch Nadal right-handed or Novak left-handed? Hi-techtennis.com does it for you. In 2007 David Nalbandian beat Nadal, Djokovic, & Federer to win a tournament, then beat Nadal and Federer the next week.**

Through 2019, US tennis experienced a steady long-term decline in per capita participation since the tennis boom of the 1970s ended. A 1984 Gallup Poll reported that one in five adults rated tennis as a top-five favorite sport to play. By 2004, the number had fallen to one in twenty adults.<sup>2</sup> Tennis courts are torn up and paved over while being invaded by pickleball and padel tennis. Conflicting instruction for beginners and return players compounded the crisis and likely contributed to the decline. If instructors can agree on proper foundational tenets from the outset, more players will want to play tennis at least recreationally during their lifetime. We must unite coaches around simplified and proven methods that drive up US per capita participation, resulting in sustained numbers and potential long-term growth.

Six key technology evolutions stood out in the 20<sup>th</sup> century: 1) women's and men's clothing, 2) the two-handed backhand, 3) conditioning/nutrition advances, 4) racket/string technology, 5) high-speed video and

analytics, and 6) internet sites/talk tennis forums. I wanted to identify which instructional evolutions stood out in the 20th century. Since the internet gave me access to coaches across the globe, I decided to research which insights or innovations stood out in the first part of the 21st century. In 2007, Coach Chuck Tomlin asked me if I knew which coaches in history were teaching to shape the ball similar to a teaching method he developed that looked at every shot as an intended "fade" or "draw." He wanted clarification on what had been taught in the past to compare to his teaching method.

While I examined what Bill Tilden and our forebearers taught, I researched deeper into how coaches taught players to hit the rising ball. After prolonged analysis, I realized over time that Tomlin's "fade and draw" paradigm might have uncovered a solution to an issue that coaches had struggled with since the 1870s. After finding no coaches, past or present, teaching this concept, I told Chuck that his method for hitting the rising ball was likely the missing link in instruction that coaches had long searched for.

## **The Seven Chapters of Volume I**

Volume 1 explains and documents how our history of instruction can help coaches, parents, and players resolve many of the current challenges presented by contradictory teaching.

Chapter 1 reveals historical information to give context as to where we lost our path and how we can better identify good instruction moving forward. We must be honest about how we got here to reverse the per capita decline of tennis participation. We must reclaim our lost and misinterpreted tennis history.

Chapter 2 details research on how very few sports build "strong bones and minds" as tennis does. Every parent should at least consider tennis a top choice for a child's athletic and overall holistic development.

Chapter 3 examines the challenge of how to apply science to a sport best played by feel and instinct. How do we know our tennis experts interpret science and data correctly? Is there a better way to analyze the Advanced Tennis Research Project spin data? Is there an overemphasis on racket head acceleration?

Chapter 4 begins a year-by-year chronology from 1874 through 1907 that adds historical context to reclaim the wisdom of our past and identify how we can improve instruction. Why were specific tips kept and others discarded? Students of tennis will find many insights from studying the 1904 entry.



Chapters 5 and 6 are written by Coach Chuck Tomlin, who introduces the Congruent Tennis Model. He developed the concept of congruent tennis after a long quest to seek better answers and resolve conflicts in instruction. It is designed to help coaches, parents, and players view and filter teaching through a higher resolution context that will help answer many of tennis' most complicated questions. Congruent tennis accommodates a wide variety of coaching styles and offers a process for assisting coaches in agreeing on the key "fundamentals." Updating and better defining old and new tennis vocabulary will help build a bridge toward coaches. Appendix G is a proposed Glossary of Tennis Definitions to Resolve Contradictions.

Chapter 7 discusses what I learned from studying differing methods. I challenge our tennis institutions to do better. Parents, players, and coaches are offered suggestions to learn from my long journey to find and identify a tennis model that works for the greatest percentage of students. The Appendix follows Chapter 7.

Tennis, then called lawn tennis, in the three decades leading up to 1907, was blessed by gifted writers who were highly educated and accomplished outside of tennis. Sarah Whittelsey was the first woman to graduate from Harvard and Yale. Her 1894 book, **A Lawn Tennis Manual**, is better than much instruction a century later. I quoted many passages from tennis authors to give the reader a sense of their exact words. All direct quotes from other writers and coaches are separated by a separate font in paragraphs and block quotes to provide a sense of reading their perspective and to differentiate from my comments. A large bibliography of footnotes for this book is posted online at "[Congruenttennis.com/Volume1Footnotes](http://Congruenttennis.com/Volume1Footnotes)."

In 1970, at age 11, **Ed Faulkner's Tennis: How to Play It, How to Teach It**, a highly acclaimed book, became my first coach. Over the next three decades, I read over five hundred tennis books documenting pages of golden tips that I hoped would improve my tennis game. I had the impression from those books and the most popular tennis writers that all excellent instruction began with Bill Tilden in the 1920s.

In 2005, I initially started the timeline with the three instructional books by Bill Tilden considered "masterpieces" of coaching; his 1920 book, **The Art of Lawn Tennis**, his 1925 book, **Match Play and the Spin of the Ball**, and his last book, written in 1950, **How to Play Better Tennis**. At that time, I was unaware of any other essential teaching books written besides Tilden's before World War II; thus, I planned to cover instruction from 1950 to 2005. I did not anticipate the ever-expanding scope of this project, which resulted in eighteen years of research, testing, editing, and seeking objective feedback for my conclusions.

For a decade, without success, I tried to get the USTA, PTR, and USPTA to review my research and findings. In 2021 USPTA President John Embree was gracious enough to grant me a lengthy interview explaining the challenges he foresaw in coaching. He provided insight into the agreed-upon five-year truce, which permitted the USTA to conduct research before stepping into coaching certification. The USTA and PTR ignored numerous requests I made to review excerpts of this book to ensure I accurately represented their views. What does history say about how they handled previous inquiries from grassroots coaches? I offered many names mentioned in this book the chance to preview excerpts before publication. That offer is extended to subsequent writings (including other editions) to anyone mentioned in this historical timeline to ensure the accuracy of their views.

## **The First Book That Claimed to Teach Modern Tennis**

Two incidents resulted in my chronicling 150 years of instructional history and following the evidence wherever it led. First, in late 2005, I did a library search for the phrase "modern tennis" to ensure I reviewed books on the hot coaching cliché of that time. The St. Louis Library was about to discard a 1904 book, **Modern Lawn Tennis**, by P.A. Vaile. The book was a revelation because the author displayed an understanding of technique that explained how today's ATP and WTA players hit their forehand. Vaile introduced a concept I did not understand initially; to hit "*forwardly and upwardly in an oblique direction across the intended line of flight of the ball.*"<sup>3</sup> For good reasons, P.A. Vaile gets much ink in this book.



## Chapter 2

### Did Ski Instructors Teach Tennis Better Than Tennis Coaches?

*"Who are you going to believe, me or your own eyes?"*  
—Groucho Marx

#### An "Inner Game of Tennis" Clinic Yields Coaching Insights

Thomas Kuhn's theory of innovation generated from outside industry experts came to mind when I examined coaching guru Jenny Rogers' story of how ski instructors got better on-court results than certified tennis coaches. England's Jenny Rogers coaches CEOs and executives in almost every field. Her famous manual, **Coaching Skills, A Handbook**, now in its fourth edition, is still considered the definitive work on coaching skills. Rogers used a tennis instruction experiment as the background to explain why too often, coaches are seduced by their own "expertise."

Timothy Gallwey coached tennis in Seaside, California, in 1971. He used the most common delivery method of most coaches, the traditional "command and control" instruction method of "do this" and "do that." Tired of talking one day, his mind wandered as he hit balls. He observed something different that happened when he shut up; his students began playing much better than he could have imagined. After watching this over time, Gallwey, in 1974, wrote the best-selling tennis instructional book of all time, **The Inner Game of Tennis**. Gallwey subsequently wrote "Inner Game" books on golf, music, and skiing. He introduced a paradigm shift in tennis instruction which focused on enhancing the student's awareness of what was happening with the ball, the racket, and the student's own body.<sup>109</sup> He developed a series of questions and instructions to quiet the student's inner mind that many coaches later incorporated through the teaching philosophy known as "guided discovery."<sup>110</sup>

In **Coaching Skills, A Handbook**, Jenny Rogers quoted a story in which John Whitmore observed in his book, **Coaching Performance**, that ski instructors could teach tennis well without knowing anything about tennis. She quoted Whitmore:

*Several of our Inner Tennis courses were so overbooked that we ran out of trained Inner Tennis coaches. We brought in two inner ski coaches, dressed them in tennis coach's uniforms, put a racket under their arms, and let them loose with the promise they would not use the racket under any circumstance. Not entirely to our surprise, the coaching job they did was largely indistinguishable from that of tennis-playing colleagues. However, on a couple of notable occasions, they actually did better. Upon reflection, the reason became clear. The tennis coaches were seeing the tennis students in terms of their technical faults; the ski coaches who could not recognize such faults, saw the participants in terms of the efficiency with which they used their bodies. Body inefficiency stems from self-doubt and inadequate body awareness. The ski coaches, having to rely on the participants' own self-diagnosis, were therefore tackling problems at cause, whereas the tennis coaches were only tackling the symptom, the technical fault. This obliged us to do more training with the tennis coaches to enable them to detach themselves more effectively from their expertise.<sup>111</sup>*

I winced when I read this story because its implications meant one of two things. Either US tennis coaches in 1996 were that bad at teaching, or those coaches' certifications weren't worth the paper they were printed on. What stood out to me was how Whitmore, trained by Gallwey, was willing to admit that the Inner Tennis coaches felt "obliged" to do more training to learn from the on-court experience of



# Chapter 3

## To What Extent Can Science Help Coaches Understand Good Form and How to Improve Instincts

*“A very great deal more truth can become known than can be proven.”*

—Richard Feynman, Nobel Prize winner in Physics

### Scientific Claims Versus Outcomes On-Court: Points of Intersection

Tennis coaches often cite the latest science. Science is the most robust method for figuring out what can be proven, but it is also very fragile and subject to the influence of coaches and researchers who have preconceived ideas. Science does not define truth; it is a constant search for a higher degree of understanding. The phrase, “the eyes cannot see what the mind does not know,” often comes into play when applying science to tennis. In an important and underrated 2017 book, **Muscle Memory Imagery: Better Tennis**, Dr. Archie Dan Smith quoted historian Yuval Noah Harari, “*Scientists usually assume that no theory is 100 percent correct. Consequently, truth is a poor test for knowledge. The real test is utility. A theory that enables us to do new things constitutes knowledge.*”<sup>143</sup>

When it comes to tennis instruction, outcomes matter. Any methods promoted by the tennis hierarchies should result in more players wanting to play tennis for a lifetime. Having fun can only take you so far with beginners. They must experience the joy of finding and building their natural athleticism. The best intentions mean nothing if they do not result in good outcomes. Positive outcomes on the court explain why Oscar Wegner became an internationally renowned tennis teacher despite many coaches resisting him, especially in America. Wegner built a simplistic way, through reverse engineering of the best strokes, for students to experience how it feels to play like a pro, even if it meant starting at slower speeds.

There are things we must accept as accurate or trustworthy, even if they can’t be proven.<sup>143</sup> The challenge is how to apply science to a sport that, since its 1874 inception, top players and coaches agreed is best played by feel and instinct and not by conscious thinking while in motion. Coaches in the early 2000s claimed that the “science” proved the ball stayed on the strings for three to four milliseconds; thus, claims that players “felt” the ball move on the strings were illusory. By 2020, the importance of feel and instinct was finally accepted by most coaches, probably unaware it was previously emphasized a century earlier. What changed the minds of those who previously cited the “science?”

Two primary factors may have convinced many establishment coaches to acknowledge the importance of “feel” and “instinct.” First, a school of grassroots coaches, including myself, stated publicly on online tennis forums that the three to four-millisecond measurement was irrelevant. We claimed the “dwell time” of the ball on the strings might feel much longer to a player than what scientists could measure. Many of us started posting quotes by tournament winners who often mentioned their feel of the ball as a primary factor in how well they played.<sup>144</sup> Vic Braden and other tennis researchers, shortly after they were challenged, spoke with pro players about the issue and later conceded that science may have missed something.<sup>145</sup>

A second factor that helped move coaches towards respecting the importance of feel and instinct was research by Malcolm Gladwell, an academic who decided to study why certain people succeeded in their endeavors. In Gladwell’s 2007 book, **Blink: The Power of Thinking Without Thinking**, he concluded that successful people often make subconscious snap decisions for reasons that were not always apparent to the athlete or the observer or in the case of Vic Braden, a tennis coach.<sup>146</sup> Gladwell confirmed Feynman’s claim that there are things we know regardless of whether science can explain them, such as how the unconscious mind rules in high-performance athleticism, and tennis was no exception. In other words, the best decisions are often those that we find impossible to explain to others.



Gladwell explained how high-level athletes made snap decisions using their unconscious minds through what is defined as thin-slicing. He interviewed Vic Braden, whom I studied devoutly for a decade. Vic spent a fortune and much of his adult life researching how to determine what made great tennis players. Braden was baffled by his acquired skill to tell with near certainty when a player was about to double fault, but he could not figure out how he did it. Gladwell pointed out how some of Braden's cutting-edge research, though necessary, was sometimes off-target. Braden and many other scientists and videographers may have aimed for the wrong target, namely highly developed players. As Wayne Bryan's 2012 letter implied, they might have served tennis better if they looked at the developmental coaches with the skillset to set the foundation for greatness.<sup>147</sup>



**Nadal drags the racket into contact before change of direction to hit up, through, and across the ball. If his arm is straight at contact, how much extension is he really using? Notice the diagonal plane?**

Gladwell warned when surveying the best in their fields, *"When we ask people to explain their thinking—particularly thinking that comes from the unconscious, we need to be careful in how we interpret their answers."*<sup>148</sup> He claimed that behavior modeling does not always work as well as advertised; for example, hitting with a great player like Pete Sampras or watching him play doesn't necessarily rub off easily. A different skill set was needed to coach a player to play tennis or any other sport well. Gladwell wrote how Braden was shocked to discover that when he surveyed champions, hardly one of them could describe how he or she hit their forehands.<sup>149</sup> By 1907, coaches often observed this phenomenon, often citing it as why they felt compelled to write books on instruction. Gladwell provided evidence that explained why expert players usually do not make great coaches, justifying the adage, *"Those who can't do, teach."*<sup>150</sup>

## **The Role of the Wrist in Tennis Strokes – Do We Understand its Function?**

Braden discussed the role of the wrist with Gladwell, *"Out of all the research that we've done with top players, we haven't found a single player who is consistent in knowing and explaining what he does...or they have answers that are simply not meaningful. Almost every pro in the world says that he uses his wrist to roll the racket over the ball when he hits a forehand."*<sup>151</sup> Braden was surprised that players cognitively did not know how to explain what they did. He showed Gladwell digitized slow-motion video. If Braden had looked at the last photo in this chapter with P.A Vaile's comments, he might have understood better when he continued,

*Why? What are they seeing? Look, see when he hits the ball? We can tell with digitized imaging whether a wrist turns an eighth of a degree. But players almost never move their wrist at all. Look how fixed it is. He doesn't move his wrist until long after the ball is hit. He thinks he's moving it at impact, but he's actually not moving it until long after impact. How can so many people be fooled? People are going to coaches and paying hundreds of dollars to be taught how to roll their wrist over the ball, and all that's happening is that the number of injuries to the arm is exploding.*<sup>152</sup>

## **The Coach Who Applied Time & Motion Principles to Tennis Biomechanics**

Often the truth is known after the fact. In the 1960s, Vic Braden began thinking about how to understand the tennis swing through science. In the 1970s, he teamed up with experts in biomechanics. Was Vic the first to do so? In Chapter 1, I pointed out how Coach Dick Bradlee and his emphasis on open stance may have at least resulted in an indirect effect on the Williams Sisters. Bradlee's place in tennis history was "ghosted" (erased) until I revived him on online tennis forums. The timeline revealed that Dick Bradlee, in the 1950s, became the first tennis coach I found to team with the world's then-best-known motion scientists to study tennis biomechanics. Bradlee's work resulted in a revolutionary tennis book that was described by



# Chapter 4

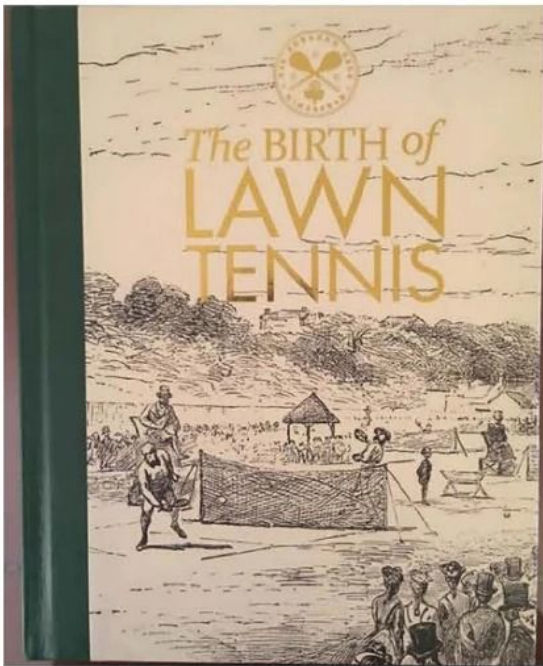
## The History of Tennis Instruction Timeline

*"The farther backward you can look, the farther forward you are likely to see."*

- Winston Churchill

### Two Important Books Capture the Significance of Tennis History

The first significant book I recommend tells the true chronological story of the invention in 1874 and the early years of Lawn Tennis. The late rare book dealer Larry Lawrence called it the best-researched book on the early history of any sport. I concur. On August 27, 2018, the first day of the US Open, I opened the online sports section of *The New York Times* and found an article about Richard Hillway and Robert Everitt's new book, **The Birth of Lawn Tennis: From the Origins of the Game to the First Championship at Wimbledon**.<sup>203</sup> Each time Richard Hillway wrote an article on tennis history (at last count, over 125), it sent me back to my timeline for fact-checking and research. Hillway consistently proved that a lot of accepted tennis history needed updating.



After the death of Bud Collins, Richard Hillway inherited the title of tennis' most eminent tennis historian. **The Birth of Lawn Tennis** helped me fill in significant historical narratives. The book is likely not only the best tennis coffee table book ever written, but it is the definitive book on the invention of lawn tennis during the 1870s. While turning the pages, I imagined walking back in time and witnessing the evolution of the game of lawn tennis. The many beautiful photos the authors chose for the book painted a tapestry of a story that is not just about the sport. **The Birth of Lawn Tennis**, which weighs seven pounds, is the ultimate gift for the tennis fans who think they have everything. Every library and tennis club should have it. It was praised universally for good reasons, including correcting false tennis history. If any book gives the sense of touring a history museum, this one does. It was published through the generosity of the All England Lawn Tennis Club.

Richard Hillway is the kind of researcher who acquired a copy of Bill Tilden's will, the original funeral book of attendees, and over 150 of Tilden's letters which demonstrated Frank Deford's 1976 biography of Tilden's career, including his last years, contained much false information. Everitt and Hillway showed that much of the early tennis history in media is inaccurate. Hillway, who owns one of the largest collections of tennis books and magazines on earth, also discovered a number of errors in the book, **The Bud Collin's History of Tennis: An Authoritative Encyclopedia and Record**. With the help of American historians such as George Alexander and Frank Phelps, both deceased, and American researcher Geoff Felder, Hillway examined travel records and discovered that countless articles, online accounts, and books that claimed Mary Outerbridge was the first to bring tennis to America in 1874 are indeed likely false.<sup>204</sup>



Until **The Birth of Lawn Tennis** was published, everything I ever read claimed Mary Outerbridge was the first person to bring lawn tennis, via a trip to Bermuda, to the United States at Staten Island on February 2, 1874. The problem is that Major Wingfield did not offer his new game of lawn tennis to the public until at least three weeks later, after receiving a provisional patent for a portable court on February 23, 1874. The game could only have spread to Bermuda after it was invented and introduced to the public. It was not until 1973, a century later, that George Alexander discovered Wingfield's first rule book on lawn tennis was published on February 25, 1874. This was the critical piece of evidence discounting Mary Outerbridge's story that she brought tennis to America in late January or early February of 1874.<sup>205</sup>

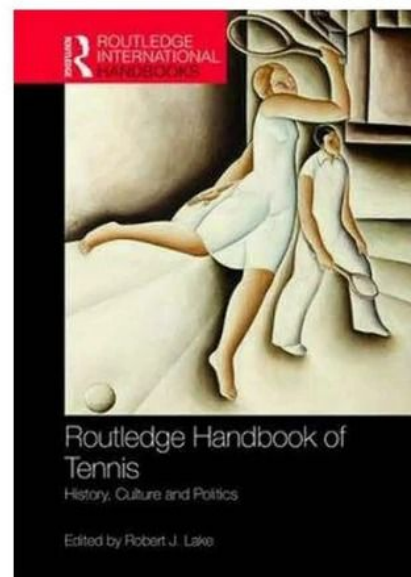
We are unsure when tennis was first played in America, but Hillway and Everitt noted that a military wife, Ella Bailey, played lawn tennis in Arizona in October 1874. Robert Minturn of New England bought a lawn tennis set in London in September 1874, but we do not know when he first used it. James Dwight and Fred Sears played tennis, most likely in America in 1875. Sometimes Dwight claimed 1874, other times 1875. Tennis was played in Westchester, New York, in 1876. Hillway and his colleagues showed that Mary Outerbridge probably brought tennis to the Staten Island Cricket and Base Ball Club in the Spring of 1877, where at least by Lady's Day in June of that year, women played lawn tennis at the club.<sup>206</sup> My independent research agreed with Hillway in that I found no evidence of tennis played in Staten Island before 1877.

Historical documentation must be as accurate as possible. Numerous prominent websites, recent historical tennis books, and academic papers that Hillway looked at are still full of errors and need to be revised regarding tennis history. Hillway and Everitt showed that both recent and early media accounts contained inaccuracies. Their book is backed up with much contemporary written evidence from the nineteenth century. If anyone wants to update their websites or check their facts in the future, check it against **The Birth of Lawn Tennis**. Hillway read the final draft eight times to ensure accuracy before publication. It is the definitive history of early lawn tennis.

After I read Hillway and Everitt's book, I had to make numerous corrections to this book, prompting me to contact Hillway. After he read my timeline, which I thought had accurate footnotes, he called me to tell me to avoid trusting internet tennis history books and coaching narratives. Hillway proved that an 1889 book by Herbert Wilberforce titled **Lawn Tennis: With a Chapter for Ladies by Mrs. Hillyard** had many errors despite being published fifteen years after the sport was invented, forcing me to make corrections.<sup>207</sup> I was grateful that Richard Hillway signed on as my historical editor after he finished a few projects.

**The Routledge Handbook of Tennis: History, Culture, and Politics** is the second significant book I recommend because of its cultural and historical background of tennis worldwide. Sports historian Robert J. Lake collected and edited a panorama of tennis articles by many authors. From the inside cover:

*Tennis is one of the world's most popular sports, as levels of participation and spectatorship demonstrate. Moreover, tennis has always been one of the world's most significant sports, expressing crucial fractures of social class, gender, sexuality, race and ethnicity – both on and off-court. This is the first book to undertake a survey of the historical and socio-cultural sweep of tennis, exploring key themes from governance, development and social inclusion to national identity and the role of the media. It is presented in three parts: historical developments; culture and representations; and politics and social issues, and features contributions by leading tennis scholars from North America, Europe, Asia and Australia. The most authoritative book published to date on the history, culture and politics of tennis, this is an essential reference for any course or program examining the history, sociology, politics or culture of sport.*<sup>208</sup>





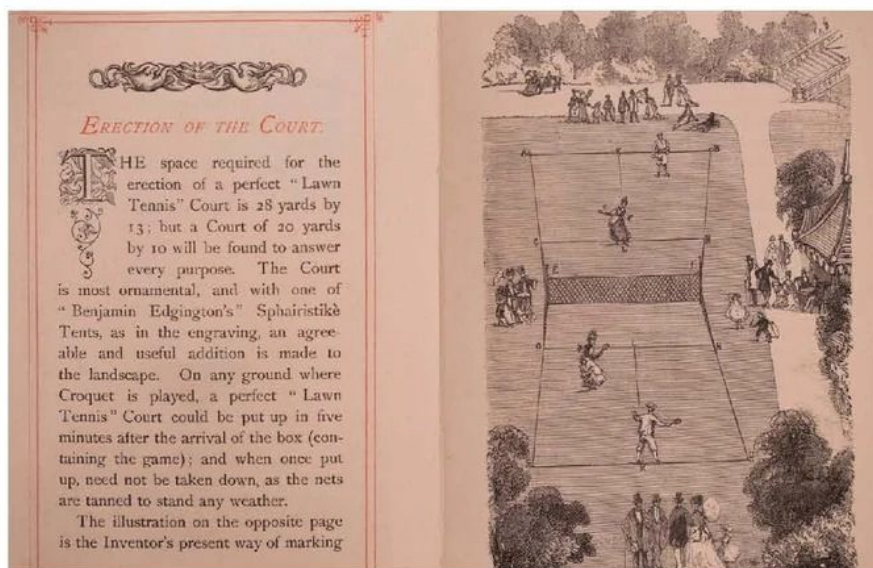
This book is a valuable addition for people who appreciate the context of sports in understanding culture and history. Lake and his collaborators wrote about how tennis often tells us a lot about the development of a society. The book has forty-plus articles that document the importance of how tennis was viewed worldwide and offers a panoramic landscape of societal evolution as tennis moved from the sport of aristocrats to all classes of people across the globe. In its earliest days, tennis was recognized for its inherent importance in enhancing good character and values in its participants, regardless of economic privilege or background. Lake and his assistant editor Carol Osborne deserve plaudits for this fascinating work described by the quote, *"This book is an edited collection of chapters authored by many of the leading, contemporary, scholarly writers on tennis in the social sciences."*<sup>209</sup>

Robert J. Lake's excellent research was invaluable, and he gave constructive criticism of the early stages of my book. Lake's first book, **A Social History of Tennis in Britain**, won the 2015 Lord Aberdare Literary Prize from the British Society for Sports History.<sup>210</sup> His books filled in context giving evidence that the evolution of British coaching paralleled certain developments in America, and I used key material from it. Hillway and Lake finally ran into each other at the Wimbledon Library in 2019 and had a chance to talk. I am indebted to both of them for their high degree of scholarship. The following pages in this chapter represent the timeline of tennis instruction from 1874 to 1907.

**1874:** The summary in this paragraph is taken from Hillway and Everitt's book, **The Birth of Lawn Tennis**. Major Wingfield patented his hourglass portable court in February of 1874 and sold lawn tennis sets to be set up on any flat surface, but especially on grass. The recent improvement of hollow vulcanized rubber balls, around the year 1870, that bounced higher on the grass made the creation of lawn tennis possible. It contributed to greater enjoyment of the game. Croquet had been popular then because men and women looked for recreational things they could enjoy doing together. The aristocratic men of England played the sport of cricket, while men and women played croquet on large smooth grass lawns.

Due to the vast reach of the British Empire extending to Australia, India, Canada and Africa, lawn tennis quickly took off with the English-speaking aristocracy, then gained steam initially with the wealthy in America. Few anticipated how the middle and lower classes would also love tennis.

Tennis historian Ben Jones wrote extensively about how even in England, tennis evolved into a popular sport of the middle class by 1901.<sup>211</sup> Tennis offered an adrenaline-creating activity for men and women to be teammates. Robert Lake wrote that a game of mixed doubles was *"the most fashionable pastime for young ladies and gentlemen seeking the intimate social pleasures of outdoor recreation."*<sup>212</sup>



**"Erection of the Court"** from Major Wingfield's second 1874 rule book. Net height at posts was initially 5 feet. The first Wimbledon in 1877 lowered the middle of the net from 4 feet to 3 feet 3 inches resulting in an emphasis on topspin. Service boxes were in the back of the court until May 1875.

**1875:** Babolat started as a company that made lawn tennis strings out of sheep gut, used in real tennis rackets since the 1500s.<sup>213</sup> Strings were traditionally called "catgut" but were never made from cats. Cow gut took over for sheep gut in the 1960s. By 1875, the sport invented initially by Major Wingfield as Lawn



# Chapter 5

## Improving Vocabulary with the Congruent Tennis Model (CTM)

By Chuck Tomlin

*“But if thought corrupts language, language can also corrupt thought.”*

—George Orwell, 1984

### Keeping Instruction Simple with a Higher Resolution

I spent about 10-12 years deeply studying what has become known as “Traditional” or “Classic” tennis instruction. With my demanding background of Naval Aviation, SEC College football, and Martial Arts, I came to this sport with an extremely high appreciation for sanctioned methods of instruction. I want to thank the many coaches who shared their efforts via books and articles for all to read. I make this point so you can realize how I began this journey with classic instruction sources such as Jack Kramer and Robert Lansdorp. I studied their teaching style and used it diligently for many years before realizing something very important was missing.

With a technical background as a Crew Resource Management (CRM) Instructor at the United States Naval Test Pilot School, you might expect complexity in my explanations; however, I believe in keeping things simple and actionable, especially under stress. My military flight experience confirmed this approach over and over. Hopefully, this method will resonate with you as well.

Remember that one of the major challenges in coaching tennis is managing contradictory instruction. These contradictions can often be the result of poor terminology. Tennis is a complex 3D sport but must be primarily described in 2D terms, pictures, and videos. For this reason, I think a good coach must be able to understand the 3D nature of the game, but often find ways to teach in 2D without losing vital aspects of the game. This is an essential concept because most of our current terms and tools are not up to the 3D presentation task, and neither are the minds of most new players in the early stages of their skill development.

Throughout this section, I will describe the things I have learned, experienced, and figured out over time that were key to helping me and other players to progress. I expect that much of what is outlined here has been seen or experienced on some level by many players. Still, this effort to explain the information for chapters 5 and 6 also led to several essential insights that you will likely find as interesting as I did during this writing process. Despite the idea that there is not a tennis swing or shape



**Federer and Nadal dragging up the slot - notice how from Slot Entry the path is very linear. Just prior to contact, the hand path curves across vs. down the target line.**



that players have not used before at some point, I think what is shared here may be the most congruent collection of tennis ideas ever presented. Some are even fresh perspectives that we have not found recorded in history.

## **Technique - The Key to Consistent Execution**

I'm not the first one to speak of "Consistent Execution" in tennis. It's probably such an obvious and common idea that it ends up hidden in plain sight, but I like to stress the importance of this phrase as the measure of everything worthwhile in tennis. This game requires an extraordinary ability to perform both open and closed skills.

The need for consistent execution is the impetus that brings us to "technique" and the terms to describe it. Some do not see the need for excellent technique, while others expect too much from technique. Often, we hear how technique is primarily about efficiency, but that is only true once effectiveness is assured. Techniques are the bio-mechanics necessary to produce certain shots consistently. We don't see players capable of high-level, consistent execution without a solid base of fundamental technique. It is important to realize the difference between the actual fundamentals of technique vs. varying degrees of amplitude in performance of fundamentals, as well as individual styles. Sound fundamentals must be present for a player to have consistent execution. If a claimed fundamental is not actually necessary for execution, then it just is not a true fundamental for that stroke,

One example of something that is often misunderstood as a fundamental is the idea of "full extension." One internet coach I viewed pushes extension as the primary key of his 3 core forehand fundamentals. He also bounces back and forth, conflating the term "fundamental" with his "observed commonalities," which further confuses what defines a fundamental. In his "Fundamentals of the Forehand" description, he goes on to cherry-pick videos to show top players using his take on "extension to the target." He points out how Nadal had more extension than the other two examples, which ironically undermines his point and instead illustrates the variability of extension even within the tiny sample size of the very best players. With any casual study of these top pro forehands, not only will the amount and direction of extension vary based on pace, but also with tactical intent and shot path. Clearly, using this version of "extension" is a variable related to the shape of the ball's flight, amplitude, or power level for a particular shot intention and is therefore not a true fundamental of the forehand. Extension is a variable factor and terms matter; especially when it comes to defining fundamentals.

On the other side of the equation, a good fundamental technique makes your power more usable but isn't as much about creating power. The way I see it, technique helps you have "access" to your power way more than it helps you to "create" power; similar to the way that good driving technique lets you use more of your car's capabilities but does not add any horsepower. It is subtle but important to place clear expectations as you work to acquire improved technique. Learning to use Roddick's serve technique can help you access more of your power, but it is probably not going to have you serving at 155 mph like Andy.

Speaking of serving like Roddick, I embraced the challenge of studying the serve for myself to see what I could deduce as the key fundamentals in order to teach it well. I then picked those players I felt were the best servers such as Sampras and Krajicek and studied their high-speed videos. I worked hard analyzing their frame-by-frame comparisons for clues to their abilities. My efforts turned out to be an enormous success, with my son becoming one of the country's fastest servers. This led me to helping several other juniors with significant serve improvements and becoming a respected serve reference on several tennis forums. This work was dubbed "The Serve Code" by some of the coaches I shared it with, and the overview is included in these chapters. This successful effort with the serve got me thinking about what could be done if I gave the same study of fundamentals to other strokes.



## Chapter 6

### Teaching Tennis Through a Higher Resolution

By Chuck Tomlin

*“Even when the experts all agree, they may well be mistaken.”*

—Thomas Sowell

#### Using an Advanced Tennis Foundation to Build Congruence

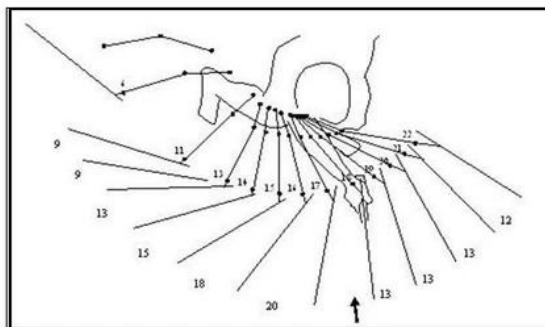
It was not my initial intention to try to unify tennis methods, but after a certain point, things did seem to naturally start falling into place. It is my nature to resolve technical conflicts by keeping contradictions to a minimum. With constant questioning of why tennis instruction lacked congruence we normally expect in other disciplines, over time better context for answers became available. Historically, as John has shown, many tennis instructional paradigms and concepts don't reconcile well at all with how the game is played by the world's best competitors. Much of the tennis instruction doesn't even match well with other published instruction. There are often big contradictions, such as how can every short ball be attackable, when at the same time, short balls are the most commonly hit groundstroke winners. Some coaches encourage lifting and even jumping during strokes, while other coaches teach players to stay down and avoid lifting the body up. Do the best players step forward to add power to their shots or do they step forward to improve their contact points? Do they use that lead foot to block and shift their momentum up or do they do it to gain forward momentum for their “weight into the ball?” Should players get their rackets back early or keep both hands on the rackets longer in front of their torsos? These and other unresolved issues often make tennis more difficult to learn.

“Match level technique” of the best players throughout history has often differed greatly from their coaching advice, as illustrated extensively in this book. Also, notice how some coaches evaluate every forehand through the lens of a specific forehand “style.” Some coaches go to the other extreme of classifying over 30 types of forehands. It is clear to me that every forehand doesn't need full extension or max shoulder turn, but there are those who confuse these “aspects of magnitude” with true “core fundamentals.” Again, these inconsistencies in this approach further demonstrate gaps and conflicts in how tennis is taught.

As I studied what the acclaimed authorities taught, the following questions arose:

- Do the major tennis organizations recognize input from the thousands of coaches in their membership or do they reserve nearly all recognition and influence for the special few?
- Does better understanding and paradigms creations come from experience in the trenches using a broad base of athletic backgrounds, or do they come from coaches with a narrow focus acclaimed by those who best peddle influence?
- Does it make sense to compare each new theory to other accepted models to see how they match up?
- Can further insight regarding other intersecting issues be put forth in a simple, understandable manner?
- Is it possible that a rejected older idea might have application when new facts are discovered?





**More proof from APAS that acceleration occurs just before impact denoted by the wider gaps just before contact. This shows how hand starts across just before contact. This is the core issue of Modern Tennis. You also see this on the Laver Serve APAS diagram in Chapter 3.**

My experience has been that when you are on the right track technically, the strategy, tactics, and even the stroke mechanics will tend to sync up to complement each other, even going so far as to illuminate aspects of one another, providing a deeper insight into the game. Many people in tennis speak of “myths,” but in Congruent Tennis we refer to them mostly as partial truths that need better explanation or context. These paradigms have been key to seeing the game in a more accurate and *integrated way* that seamlessly connects technique with strategy and tactics. We show how the different shapes of different ball flight paths can exploit the Smart Target Lanes from different parts of the court. While no model is perfect (remember- the map is not the territory), with ATFa and CTM we seek to continually update with the highest resolution congruent tennis concepts, where the instruction blends and even highlights other aspects of the game. One

of my favorite joys of working with the CTM and the Advanced Tennis Foundation are the predictable yet revealing comments from coaches and players alike. The most common ones that immediately come to mind are observations about how “things are really coming together for me now, after seeing this” or “how did I miss seeing this watching matches all these years?” Maybe the most common comment and highest praise is from excellent former ATP and WTA players and former top college players saying how much better they could have been with this Congruent type of instruction.

While CTM is something I’ve worked on for years, the Advanced Tennis Foundation (ATFa) is an additional concept that comes to us from a brilliant coach and notable author, David Smith. Dave’s contributions to the game have been both deep and broad. He is that rare instructor who has brought many players into the game as a “Developmental Coach,” and he has also coached high-performance players. He has worked in private practice, but he also worked for years in the school systems. In his writings, he covered the concept of an “Advanced Foundation” that Coach Carpenter and I agreed was simple and exceptional (two of my favorite things for tennis). The central idea behind ATFa tennis is that a coach should start all players with an “advanced foundation/” meaning their initial mechanics will not have to be later corrected. Dick Bradlee in 1958 wrote we should never teaching bad starting positions. Players should learn strokes, stances, and tactics that will provide a strong foundation to their play no matter how far they travel in this wonderful game.

## Smart Target Hitting Lanes

The best way to hit safely and a bit wider is to utilize “Smart Targets,” which are nice examples of the Congruent Tennis Perspective. There are many perspectives concerning tactics, strategies, and targets. People have long thought that tennis strategies were about targeting different parts of the court based on the coverage openings left by their opponents as well as where the balls should bounce. For this reason, players are often very distracted by small changes in an opponent’s position. The players often try to respond to their movements by changing their shots. The CTM hitting lanes are defined by what I call “Smart Targets” that have exploded and blown up that paradigm to a large extent. In reality, there are only two primary targets with a “plus one” option on a tennis court: hit left of them, hit right of them, or hit “at them” to jam them. Smart Target Tennis is a simple system to help players see and understand how two of these target lanes never move, as well as learn how to execute through these lanes more consistently.

**Rest of Chapter 6 not in these excerpts**



## Chapter 7

### How Parents, Coaches, and Players Can Navigate the Challenge of Contradictory Instruction

*"Where there is much desire to learn, there of necessity will be much arguing, much writing, many opinions; for opinion in good men is but knowledge in the making." - John Milton*

#### Lessons Learned from the 1874 to 2023 Timeline

The instructional timeline is an evolutionary lens through which we can examine what went right against what could be improved with better coaching methods. History offers insights to the best solutions to the curse of contradictory instruction that has plagued our great sport since its earliest days. Events make more sense when not viewed in isolation. Once I found Dick Bradlee, I suspected Richard Williams teaching open stances to Venus and Serena in the mid to late 1980s meant there likely was a connection to Bradlee.<sup>873</sup> The connection was through Coach Daphne Chinn. Bill Tilden wrote the tennis swing is a baseball swing.<sup>874</sup> Shortly afterward, the Beasley Method, which taught tennis through baseball swing mechanics, made Mercer Beasley famous.<sup>875</sup>

This book calls for a conversation between grassroots coaches and the establishment hierarchies to better define a simplified Advanced Tennis Foundation. Conflicting instruction builds distrust with parents and frustrates students when the new coach claims he has to clean up the previous instructor's influence. The path to mitigating contradictory instruction begins with a hierarchy of values. Though my first principle in instruction is congruency, I might label it 1A. Right behind is 1B, the principle of simplicity.

Coaches and writers can reference and understand the origins of many of today's claims. Tennis racquets had been used for centuries before Major Wingfield wrote his rules for lawn tennis in 1874. By the time Henry Jones and friends tweaked the rules to create the game we play today, William Renshaw in the 1880s created the first lawn tennis model of "good form." The best players quickly learned to avoid hitting the ball "fat" and to instead work across as they stroked through the ball using rotational force, which Coach Tomlin explained in Chapter 5 as the "cam effect." Through trial and error, players learned to bend the ball with spins into the court. Any effective shot or style was quickly tested for adoption by the next generation.



**1885 photo backs the assertion that the best players in early on played much like today's players differing only in the degree of intensity. Miss Louisa Martin, Ireland's top woman, ran around her backhand into an open stance, and despite the dress, coiled into the hip (loaded) and set up to drag the racquet from the slot as she uncoiled to hit up, though, and across the ball.**

Since tennis' earliest days, nearly all coaches agreed seeking to take time away from the opponent by hitting the rising ball was a key to winning. Dick Williams, Henri Cochet, and Fred Perry were known for taking the ball early and on the rise long ago.

Human biomechanics are finite. Louisa Martin, shown here in a full-length dress, likely knew the semi-open stance from the left sideline gave her last second options to hit an inside in or inside out forehand in the 1800s. Notice the "lag" of her racket as the ball nears.

Two ways to shape forehands quickly emerged in the 1800s, the lift topspin and the forehand drive. The forehand drive was taught as a flat shot hit with a plain face racquet and



the forehand lift was an up-and-down arcing topspin shot. Coaches differed on how much topspin was most effective. The straight-arm forehand was initially the preferred forehand style to teach, but over time the bent-arm forehand became the preferred hitting style. This might be because J. Parmly Paret taught the bent arm was easier to hit with and the great Bill Tilden hit his forehand with a bent arm.<sup>876</sup>

Should players be using both straight arm and bent arm forehands depending on their intended shot trajectory and shape? We have seen a recent spurt of pros who use the straight arm forehand. We must better explain the latest swing styles and why they may or may not develop naturally as a player advances.

This book's title was inspired by the fact that there was considerable "Modern Tennis Instruction" taught before 1915. At some point, primarily in England, coaching veered off into something now referred to as "classic or traditional" instruction which ironically departed in critical ways from how the top players have always played. The first tennis instructional writings observed that the best players hit across the ball. Instruction began to change when Bill Tilden decided that the turn, step and hit method was the best way to hit a forehand. He published his forehand theories in the "Top Ten Tennis Commandments" in *American Lawn Tennis* in 1919 even though Tilden rarely hit using that method.<sup>877</sup> Dick Bradlee claimed that Tilden's mistake in promoting the classic forehand took tennis instruction in the wrong direction.<sup>878</sup>

The growing coaching community that promotes Congruent Tennis does not seek to claim anyone is wrong. A quote by Buckminster Fuller explained why I decided the Congruent Tennis Model was the best way to move tennis instruction forward, "*you never change things by fighting the existing reality.*"<sup>879</sup> In order to change something, we must build a new tennis model that makes the existing model obsolete.

It is time to update our tennis vocabulary, particularly for beginners. It must be confusing when players have a private coach and attend a tennis camp where they practice drills with very different concepts. I sometimes send students to local academies to drill in a group setting with their peers. A large Atlanta tennis center has coaches emailing parents that their children (my students) are not following the recommended teaching methods. What makes this such an ironic problem is that my players cited by this tennis facility are usually rated at or near the top of their group's ranking ladders. If my student hits across the ball on volleys resulting in sharp-angled shots, why does the coach running the drills insist they punch and drive their



**The ball pockets in the strings to some degree. The nature of the "pocket" changes at impact. When you work across the ball, you create a more predictable response.**

volleys deep rather than respect their results? I have had countless parents ask me why other coaches do not respect the advanced tennis foundation we encourage the kids to build on.

Shadisha Robinson, a former WTA tour player, often invites me to teach clinics for her youngsters. I always enjoy working for her because she values the principle of congruency. We teach and drill the same biomechanics to beginners and intermediates to build the same advanced foundation to grow their game. Our trust in each other resulted from her experience testing the Congruent Tennis Model and offering feedback. Shadisha was humble enough to recognize many of our improved ways of teaching.

When competing at the highest level, her best shot was a high forehand drive, hammered inside-out through the court. Although she trained under many prominent coaches, she learned to hit this shot instinctively. Upon reflection, she realized that none of the coaches she worked with taught her how to hit that shot. They shouted, "great shot," or "perfect," when she hit it effectively. After 15 years of coaching, she had never been exposed to a way to teach it cognitively to players. None of her previous instruction contained the high-resolution teaching of the Shot Matching Matrix that is best understood one you understand the Fade and Draw paradigm.

## **Is Originality Making Something New out of Something that Already Exists?**

Despite all the technology and research I examined in the last twenty years, I think there were more than a few coaches pre-Tilden who would not be that impressed by the many YouTubers claiming they found



## APPENDIX B

### Coach Sydney Llewellyn's 1984 Interview from "Black Champions"

This interview transcription is from a filmed series of interviews titled "Black Champions."

Context by John B. Carpenter: This interview is an important record of tennis instructional history. I transcribed this from a one-hour video clip of Coach Sydney Llewellyn in the campus library of Washington University in St. Louis. If anyone would like to access the entire interview online, the *Black Champions* website is listed at the end of this interview.

For much of the twentieth century, black tennis players and coaches in the US were not granted equal opportunity and access to media and coaching forums. The era of segregation and Jim Crow meant that even after 1965 there was a limited market for tennis books written by black coaches and authors. It would not surprise me if there were self-published tennis instructional books passed around the black community before Daphne Chinn self-published her 1986 book, **Tough Tennis: How to Play Like a Pro**.

Tennis greats Althea Gibson and Arthur Ashe both credit Sydney Llewellyn as a primary influence. Llewellyn managed and coached Althea through the five major championships that she won as an amateur before she turned professional. Coach Llewellyn was born in 1911 in Jamaica and died in 1989. Once he made the decision to coach at age 35, he studied his craft intensely. Sydney studied boxing movements to teach how to avoid the dreaded "arm swing" that plagued tennis coaching. To help in that task, Llewellyn invented the Equiform, elastic cables with handles (pictured below) that he used to teach the body swing vs. the arm swing. After he became the first black coach certified by the USLPA (now USPTA), he focused on training coaches. I used a few pictures from Stacy Baker's article "Tennis in Harlem," published in *The NY Times*.<sup>942</sup>

I recorded the *Black Champions* interview of Llewellyn on cassette tape. I later found the video posted on the Washington University website. The transcription is edited to accurately reflect the film recording.

I wrote in Chapter 1 that Coach Llewellyn taught a system that *Time* magazine in 1957 called 'ruthless geometry.'<sup>943</sup> Llewellyn named it the Theory of Correct Returns. He noticed that players naturally hit to the open spot once a player had been moved to one side of the court. He wondered what happened if you did the opposite of hitting to the open "big target" and instead focused on angled or shorter shots that forced the opponent to move on a longer diagonal. We can learn a lot of tennis and life wisdom from Coach Sydney Llewellyn in this interview.



A 371-page 1939 book, **The Negro in Sports**, by Edwin Bancroft Henderson, is a detailed record of early black sports history. Henderson is known as the grandfather of black basketball.<sup>944</sup> The third edition published in 2014 retained the original manuscript. Chapter XI is titled "Pioneers in Tennis."



## Appendix H

### Comparison: The Greatest Tennis Player Ever and the Grand Slam Tournaments.

by Richard Hillway

#### Introduction - It Is Probably Impossible to Fairly Compare Champions from Different Eras.

Who is the greatest tennis player who ever lived? Today that person is called the GOAT, the Greatest of All-Time. A few decades ago, the goat was considered the team player who made a crucial mistake at a key moment and lost the big game. That has changed. What is the best method for determining the greatest male and female player of all time and also fairly listing in order the top 10, top 25, or top 100? Is there one system that works, one that accurately compares tennis players of different eras? Many researchers have tried to compare these players and have proposed their own systems for that purpose. I have seen a number of such proposals, but never one that fairly and accurately compared players of different eras. One of my mentors, Frank Phelps (1917-2010), America's leading tennis historian for decades and the one Bud Collins sometimes went to for answers, was not interested in comparing players because he did not think that it could ever be done accurately or fairly. I have come to this same conclusion. There are simply too many differences or variables among players of different generations. For example, once I asked Todd Martin who was CEO of the International Tennis Hall of Fame whom he considered the better player between Sampras and Federer whose careers were not many years apart and they had even competed against each other. Martin immediately answered that it was impossible to compare them because they played with different tennis balls. Players may be compared more accurately to others of their own generation, but even that is not easy to do. Nevertheless, it is fun to rate players from the past and present against each other. I hope researchers today continue to invent new ways or new systems to do this.

The purpose of this article is NOT to suggest a single method to be used in comparing players of different eras. I am not offering a plan of my own. Rather it is to make suggestions that might help others who want to try it. I will show what I consider weaknesses in some of the methods that have already been tried. I will also offer many details concerning the records of tennis champions that might help researchers in their quest to compare them. More details will be offered concerning the early players than the present ones since most readers are probably familiar with the records of recent players. Here follows a review of some of the WEAKNESSES in comparison models that we see commonly put forth by the media and by others.

If players are compared simply upon their win-loss records, no modern player could be rated above Suzanne Lenglen who lost only one match during her 8 years of international amateur play (1919-1926; Ted Tinling who knew both players, described how Lenglen would have beaten Chris Evert by hitting short angles and opening up the court "like an oyster"), or Helen Wills who lost very few matches in 17 years (1922-1938) and won 19 of 22 grand slam singles titles in her career. Or, in spite of poorer win-loss records, the modern players might be given more value in a comparison, largely because they do have so many advantages over the early players. As I wrote earlier, the purpose of this article is not for me to offer my own player comparison, but rather for me to provide you with thoughts, reasons, facts, examples and statistics which might help you to create your own models with which to compare players from different time periods. Hopefully, this will provide a change of pace in this very well researched book on the history of tennis instruction by John Carpenter, with two chapters in which Chuck Tomlin provides the very best instructional information available today in creating modern tennis champions.



Suzanne Lenglen, world champion 1919-1926, hitting a Buggy Whip Forehand.



**1. GRAND SLAM SINGLES TOURNAMENT WINS ONLY.** Many media player comparisons today are too simple. They take into account only one criterion - how many grand slam singles tournaments a player has won. If only one measure is used, this is probably the most important one since almost all the top players compete in the majors and they try their hardest in these events. But it is too simple using only one measure? If Nadal had won one more major singles title than Djokovic would that by itself prove that he was the better player? And this measure favors the modern players over the earlier ones as will be shown. What about LOSSES? Should losses be counted? I will list the wins and losses in the major tournaments by various champions and you may decide. What about wins in DOUBLES? Should Martina Navratilova's 41 doubles titles in majors be counted or totally ignored? Perhaps Davis Cup and other Cup matches should be considered, and Olympic medals, along with ATP 1000 tournaments and minor tennis tournaments. However, exhibition matches should never be counted.

**2. A number of player comparisons have been offered that include ONLY PLAYERS FROM THE OPEN ERA** which began in 1968 when finally, the amateur and professional players were able to compete against each other in open tournaments, on an equal playing field. Golf had allowed pros and amateurs to compete against each other in the same events for decades. But not tennis. Professionals were not allowed in the 4 majors until 1968, nor in any other sanctioned amateur tournaments. It was not until 1926 that an important professional tennis tour began, and from that time until 1968 the playing professionals competed against each other on head-to-head, city-to-city pro tours or in pro tournaments. Three major professional tournaments evolved - the US Professional Championships, Wembley in England and Paris in France. But since the amateurs and pros were completely separate in their matches and results, how could anyone accurately compare them. From about 1934 through 1967, almost always the top professional was a stronger player than the leading amateur. For example, Rod Laver probably would not have won any of his grand slam 4 titles in 1962 had there been open tennis since, at that time, pros such as Ken Rosewall and Lew Hoad were better players than Rod. But Laver may have won a grand slam of all 4 titles in 1965, 1966, or 1967 when he was at his peak and the best in the world. In those peak years, as a pro he was kept out of the 4 grand slam tournaments. Partly because it is so difficult to measure the pro records against those of the amateurs during that period, many have simplified things by just pretending tennis started in 1968 and when records are mentioned, they are prefaced with DURING THE OPEN ERA. This is often the case on the Tennis Channel.



Laver's left forearm was much larger than his right one. Marty Mulligan had a match point to win 1962 French Open, but Laver came back to win the match and won all 4 majors in 1962 as an amateur.

**3. MALE PLAYERS AND FEMALE PLAYERS.** Some comparisons include both male players and female players together on the same comparison lists. This is silly. I love watching women's matches on TV and in person. I think I enjoy them about as much as watching the top men compete. The top women players play at a very high level. They are powerful, fit and most have all the shots. They are terrific. But none of them approach the level of play exhibited by the male tennis players on the circuit. Men and women do not compete against each other in any of the major tournaments, and hardly ever in minor events either. Nor in football, basketball or soccer. Why not? It is because they do not play at the same level. In today's world, some males have become females and have been allowed, therefore, to compete against females at least in track and swimming. Although somewhat weaker than



## Appendix I

### “The Best Tennis Instruction Book Never Published by a Parent/Coach”

By John Carpenter, USPTA Elite Pro, Tennis Instructional Historian copyright Jan 2021

The articles written below by a famous parent/coach were somehow lost in tennis history. There are many books about Suzanne Lenglen (b.1899) of France, known for her acrobatic style as she dominated women's tennis like no one before or after her. While researching the 1928 timeline entry, I noticed Mary K. Browne mentioned in her first book that Charles Lenglen wrote an article about his daughter. Numerous biographies, articles, and internet searches never mentioned that Charles, Suzanne's sole developmental coach, published anything on how he raised his daughter to become a world champion in his own words. There were numerous interviews of Charles Lenglen quoted in tennis literature but Richard Hillway told me he was unaware that Suzanne's father had written any such article. Once I found the long article, written by Charles Lenglen, I concluded that his story of how he coached his daughter to World No. 1 as a teenager might be a worthy case study for parents and coaches today. I transcribed it verbatim so we all can benefit from the rich lessons the father's story of the most dominant player to ever play tennis can teach us.



1923 newspaper photo covered interview with Suzanne in which she mused she would love to coach 17-year old newly crowned US Champion Helen Wills. With two tennis balls in her left hand, Lenglen stepped out to load hips while setting up to rotate her torso into the shot

History views Charles as the first overbearing tennis parent. He admitted, *“I was a hard taskmaster, and although my advice was always well-intentioned, my criticisms were at times severe, and occasionally intemperate.”*<sup>947</sup> Charles Lenglen's article was very long. For the purposes of this book, I chose to transcribe the six installments published in the *Minneapolis Star Tribune* as they appeared each week.<sup>948</sup> Lenglen's articles would have made an excellent instructional book.

Alan Little, a longtime Wimbledon librarian whose 2007 bio of Lenglen is excellent, documented her amateur record at 341-7, the best of any international player in history. There were varying press reports about her birthplace, so Little searched Paris records and found out she was born in Paris. Apparently, he and the other biographers of Lenglen did not have access to these articles by her father that contained important history and facts about his daughter. Charles gave Suzanne's date of birth and where she was born.

Suzanne lacked a big serve, but she came to the net often in the middle of a rally. Observers and opponents commented she volleyed like a man. When the 5' 7" 20-year-old showed up for the Wimbledon 1919 final without a corset/girdle wearing short sleeves and a shorter skirt than anyone before her, spectators gasped as she defeated 7-time defending Wimbledon champion Dorothea Lambert Chambers in a thriller, fighting off two match points in the third set. Suzanne may have felt the pressure of being a champion going forward as her father was later observed passing her a flask with cognac in matches to help calm her nerves.

Suzanne was famous worldwide, maybe more than Bill Tilden. For eight straight years, 1919-1926, she was the acknowledged best female player in the world despite not winning a major in 1924 after missing most of the year with illness. In the summer of 1926, Suzanne declared she was joining the professional tour and promptly disappeared from all amateur rankings and tournaments, including the four majors. Her 1919 French



***“This book claims the Congruent Tennis Model offers a higher resolution to teach tennis with and I’m here to confirm that it does! This is coaching at a whole different level.”***

***— Johan Kriek 2-time Major Winner***

**There has been one lone constant throughout the history of tennis.**

**All the best players played what is referred to as Modern Tennis. It was the instruction that veered off from the proven championship form.**

**This book focuses on the Titans of Instruction that were careful to keep their teaching in alignment with how the best players played.**

**A Historical Parade of Tennis Instruction as Taught by the Masters**



**Suzanne Lenglen  
1919–1926 World Champ**



**Roger Federer World  
Champ 2004-2007 & 2009**

**In 1919, Suzanne Lenglen taught and used the same straight-arm forehand used by Federer, Nadal, and Alcaraz. Notice similarities?**

***“Coach Tomlin understands that tennis is a 3-D sport that has been previously explained in 2-D words. His gift is in his ability to organize a system and a vocabulary that removes barriers that take tennis instruction from the darkness of analog explanations into the light of high-resolution understanding.”***

**— Coach Steve Kaplan, USTA Lifetime Achievement Award Winner**

**Exclusive article comparing the Greats  
and Majors by Tennis Historian Richard Hillway,  
co-author of *The Birth of Lawn Tennis***

ISBN 979-8-9864285-0-5



9 798986 428505 >